Damage Identification of Bolt Connections in a Steel Frame

Jann N. Yang[1]; Ye Xia[2]; and Chin-Hsiung Loh[3]

Abstract: It is well-known that damage in a structure is a local phenomenon. Based on measured vibration data from sensors, the detection of a local structural damage requires the finite-element formulation for the equations of motion, so that any change in stiffness in a structural element can be identified. However, the finite-element model (FEM) of a complex structure involves a large number of degrees of freedom (DOF), which requires a large number of sensors and involves a heavy computational effort for the identification of structural damages. To overcome such a challenge, we propose the application of a reduced-order model in conjunction with a recently proposed damage detection technique, referred to as the adaptive quadratic sum-square error with unknown inputs (AQSSE-UI). Experimental data for the shake table tests of a 1/3-scale six-story steel frame structure, in which damages of the joints were simulated by loosening the connection bolts, have been available recently. Based on these experimental data, it is demonstrated that the proposed combination of the reduced-order finite-element model and the adaptive quadratic sum-square error with unknown inputs is quite effective for the damage assessment of joints in the frame structure. **DOI: 10.1061/(ASCE)ST.1943-541X.0000831.** *© 2013 American Society of Civil Engineers.*

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Introduction

The ability to detect damage using vibration data measured from sensors will ensure the reliability and safety of structures. Hence, analysis methodologies for structural damage identifications based on measured vibration data have received considerable attention recently (e.g., Doebling et al. 1998; Alvin et al. 2003; Bernal et al. 2004; Chang 2005). Time-domain analysis techniques have been studied and developed to quantitatively determine structural damage, such as a reduction of stiffness, including the least-square estimation (LSE) (e.g., Loh et al. 2000; Lin et al. 2001; Smyth et al. 2002; Yang and Lin 2004; Yang et al. 2007b), the extended Kalman filter (EKF) (e.g., Hoshiya and Saito 1984; Shinozuka and Ghanem 1995; Sato et al. 2001; Yang et al. 2006, 2007a; Zhou et al. 2008), the Monte Carlo filter (Yoshida and Sato 2002; Tanaka and Sato 2004), the sequential prediction error method (Goodwin and Sin 1984; Yun et al. 1997), the sequential nonlinear least-squares estimation (Yang and Huang 2006, 2007; Huang and Yang 2008), and others (e.g., Ching et al. 2006; Wu and Smyth 2007; Lin et al. 2005).

Since damage in a structure is a local phenomenon, the finite-element formulation for the equations of motion is required for structural damage identifications (Yang et al. 2009). Based on the finite-element formulation, a structural damage identification method that uses only a limited number of sensor measurements has been developed recently (Huang et al. 2010). However, the finite-element model of a complex structure involves a large number of degrees of freedom (DOF) so that (1) a heavy computational effort is needed to detect the damage, and (2) a large number of sensors are required. For practical applications, it is not possible to install sensors in the health monitoring system to measure either all the external excitations (inputs) or the acceleration responses (outputs) at all DOF. In fact, some external loads, such as wind and traffic loads, are not measurable and some response quantities, such as the rotational accelerations, are difficult to measure if not impossible. It is also highly desirable to install as few sensors as possible in the structural health monitoring system. To overcome these challenges, the use of the reduced-order finite-element formulation based on the method of static condensation (Chopra 2001) is proposed.

For steel frame structures, the stiffness of the connection joints has a significant influence on the dynamic behavior and integrity of the entire structure (Jaspart 2000; Nethercot 2000). Hence, one type of damage in the steel frame is the loosening of bolt joints, which results in a reduction of the rotational stiffness of the joints. Damage identification of loosened bolt joints have been investigated in Weng et al. (2009), and the same problem will be studied herein. In Weng et al. (2009), a method of damage identification was proposed based on a combination of (1) the subspace identification (SSI) technique (Van Overschee and De Moor 1996; Mevel and Basseville 2003), and (2) a progressive finite-element model updating and a large-scale optimization using a nonlinear least-squares approach. For this approach, the SSI technique is applied first to obtain the modal properties of the structure, followed by the optimization procedures in the time-domain to determine the unknown structural parameters numerically. The drawbacks of such a damage identification technique include the requirements of measuring (1) acceleration responses at all DOFs of the structure, and (2) all external excitations. Likewise, the numerical optimization procedures are quite time consuming, in particular when the number of unknown stiffness parameters is large.

In this paper, we propose a damage identification method based on the use of the adaptive quadratic sum-square error with unknown inputs (AQSSE-UI) recently developed (Huang et al. 2010) in conjunction with the reduced-order finite-element formulation to identify the loosening of bolt joints in frame structures.

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The loosening of the bolts at a joint considered herein indicates a reduction of rotational stiffness at the connection. While other damage mechanisms that induce yielding and inelasticity at joints during strong earthquakes are other important damage modes, in which nonlinear structural models are needed, this type of nonlinear damage is beyond the scope of this paper. Thus the method presented herein is limited to situations where the primary structure can still be treated as a linear system.

The advantages of the AQSSE-UI technique are (1) the analytical recursive solution for the determination of unknown structural parameters is available, (2) the external excitations do not need to be measured, and (3) only a limited number of measurements for the acceleration responses are needed. Although the AQSSE-UI damage detection technique has been proposed previously (Huang et al. 2010), it was demonstrated using numerical simulation results and a simple small-scale test specimen. In this paper, experimental test results using a large-scale six-story steel frame are used to demonstrate the use of AQSSE-UI technique in conjunction with the reduced-order finite-element model. While the reduced-order finite-element model is well-known in structural dynamics, its application to the damage detection of structures is significant as explained in the following. The full-order finite-element model involves many independent variables, some of which are either different or impractical to measure, such as the rotational acceleration of a joint. When the full-order finite-element equations of motion are used for the damage detection, the recursive solution will not converge without the measurements of rotational accelerations at some joints. This is true for all damage detection techniques, such as the extended Kalman filter. Through the condensation of the rotational degrees of freedom, the recursive solution converges nicely without the need of measuring rotational accelerations. Thus, the use of the reduced-order finite-element formulation eliminates the undesirable measurements and also reduces the number of sensors required. The capability of the proposed method in detecting the damage of joint connections in a structure will be demonstrated using the shake table experimental data for a six-story steel frame, in which various joint damage scenarios have been introduced by loosening the bolts.

Adaptive Quadratic Sum-Square Error with Unknown Inputs

Equation of Motion Based on Reduced-Order Finite-Element Model

Based on the finite-element formulation, the vector equation of motion of an s-DOF complex linear structure can be expressed as

$$\underline{\mathbf{M}}\,\ddot{\underline{\mathbf{x}}}(t) + \underline{\mathbf{C}}\,\dot{\underline{\mathbf{x}}}(t) + \underline{\mathbf{K}}\,\underline{\mathbf{x}}(t) = \underline{\eta}\,\underline{\mathbf{f}}(t) \tag{1}$$

in which $\underline{\mathbf{x}}(t) = [\underline{x}_1, \underline{x}_2, \ldots \underline{x}_s]^T = s$-displacement vector; $\underline{\mathbf{M}} = (s \times s)$ mass matrix; $\underline{\mathbf{C}} = (s \times s)$ damping matrix; $\underline{\mathbf{K}} = (s \times s)$ stiffness matrix; $\underline{\boldsymbol{\eta}} = [s \times (\bar{m} + r)]$ excitation influence matrix; and $\mathbf{f}(t) = (\bar{m} + r)$-excitation vector. The dimension s of the finite-element formulation in Eq. (1) is very large, so that (1) the computational efforts for the determination of all unknown parameters, such as $\underline{\mathbf{C}}$ and $\underline{\mathbf{K}}$, are quite involved, and (2) the required number of sensors is large. To overcome these challenges, the number of equations of motion in Eq. (1) are reduced using the static condensation method as follows (e.g., Chopra 2001). The displacement vector $\underline{\mathbf{x}}(t) = [\underline{x}_1, \underline{x}_2, \ldots \underline{x}_s]^T$ is divided into two vectors, $\mathbf{x}(t)$ and $\mathbf{x}^*(t)$, where $\mathbf{x}(t) = [x_1, x_2, \ldots, x_m]^T$ is an m-dimensional displacement vector representing the primary degrees of freedom system, and $\mathbf{x}^*(t) = [x_1^*, x_2^*, \ldots, x_{s-m}^*]^T$ denotes the secondary degrees of freedom to be condensed. Based on the method of static condensation, Eq. (1) can be condensed into an m-DOF system as follows:

$$\mathbf{M}(\theta)\ddot{\mathbf{x}}(t) + \mathbf{C}(\theta)\dot{\mathbf{x}}(t) + \mathbf{K}(\theta)\mathbf{x}(t) = \boldsymbol{\eta}_1\underline{\mathbf{f}}(t) \tag{2}$$

in which $\mathbf{M}(\boldsymbol{\theta})$, $\mathbf{C}(\boldsymbol{\theta})$ and $\mathbf{K}(\boldsymbol{\theta})$ are $(m \times m)$ condensed mass, damping and stiffness matrices, respectively, and $\boldsymbol{\eta}_1$ is a $[m \times (\bar{m} + r)]$ excitation influence matrix. In Eq. (2), $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_n]^T$ is an n-unknown parametric vector with θ_i $(i = 1, 2, \ldots, n)$ being the ith unknown parameter of the entire structure, including damping, stiffness, and other parameters of the original finite-element system in Eq. (1). Further, the $(\overline{m} + r)$-excitation vector $\underline{\mathbf{f}}(t)$ is divided into two vectors, denoted by $\mathbf{f}^*(t)$ and $\mathbf{f}(t)$, respectively, where $\mathbf{f}^*(t) = r$-unknown (or unmeasured) excitation vector and $\mathbf{f}(t) = \overline{m}$-known (or measured) excitation vector. Then, Eq. (2) can be expressed as

$$\mathbf{M}(\boldsymbol{\theta})\ddot{\mathbf{x}}(t) + \mathbf{C}(\boldsymbol{\theta})\dot{\mathbf{x}}(t) + \mathbf{K}(\boldsymbol{\theta})\mathbf{x}(t) = \boldsymbol{\eta}^*\mathbf{f}^*(t) + \boldsymbol{\eta}\mathbf{f}(t) \tag{3}$$

in which $\boldsymbol{\eta}^*$ is a $(m \times r)$ excitation influence matrix for $\mathbf{f}^*(t)$, and $\boldsymbol{\eta}$ is a $(m \times \overline{m})$ excitation influence matrix for $\mathbf{f}(t)$. Note that Eq. (3) is a general equation in which $\overline{m} = 0$ if no excitation is measured and $r = 0$ if all excitations are measured. In what follows, the bold face letter represents either a vector or a matrix.

Introducing a state vector with a dimension of $2m$, i.e., $\mathbf{X}(t) = [\mathbf{x}^T, \dot{\mathbf{x}}^T]^T$, one can transform the equation of motion in Eq. (3) into a state-space equation

$$\frac{d\mathbf{X}(t)}{dt} = \mathbf{g}(\mathbf{X}, \boldsymbol{\theta}, \mathbf{f}^*, \mathbf{f}) + \mathbf{w}(t) \tag{4}$$

in which $\mathbf{w}(t)$ is an added model noise vector with zero mean and a covariance matrix $\mathbf{Q}(t)$, representing the uncertainty of the finite-element model.

Adaptive Quadratic Sum-Square Error with Unknown Inputs (AQSSE-UI) Approach

A recently proposed structural damage detection technique (Yang et al. 2009; Huang et al. 2010), referred to as the adaptive quadratic sum-square error with unknown inputs (AQSSE-UI), will be used for the detection of joint damages in a frame structure. In this approach, the state vector, $\mathbf{X} = [\mathbf{x}^T, \dot{\mathbf{x}}^T]^T$, is considered as an implicit function of the unknown parametric vector $\boldsymbol{\theta}$, i.e., $\mathbf{X} = \mathbf{X}(\boldsymbol{\theta})$. To determine the unknown parametric vector $\boldsymbol{\theta}$ and unknown excitation vector $\mathbf{f}^*(t)$, a l-dimensional response vector $\mathbf{y}(t)$ is measured, referred to as the observation vector, which can be expressed in a nonlinear discrete equation (at $t = i\Delta t$) as follows:

$$\mathbf{y}_i = \mathbf{h}[\mathbf{X}_i(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i, \mathbf{f}_i^*, \mathbf{f}_i] + \mathbf{v}_i \tag{5}$$

in which $\mathbf{y}_i$ is an l-dimensional observation (measured) vector; $\mathbf{f}_i^*$ is an unmeasured r-dimensional excitation vector; $\mathbf{f}_i$ is a measured $\overline{m}$-dimensional excitation vector; and $\boldsymbol{\theta}_i$ is the unknown parametric vector at $t = i\Delta t$ (sampling time step Δt), respectively, i.e., $\mathbf{y}_i = \mathbf{y}(t = i\Delta t)$, $\mathbf{X}_i = \mathbf{X}(t = i\Delta t)$, $\mathbf{f}_i^* = \mathbf{f}^*(t = i\Delta t)$, $\mathbf{f}_i = \mathbf{f}(t = i\Delta t)$ and $\boldsymbol{\theta}_i = \boldsymbol{\theta}(t = i\Delta t)$, In Eq. (5), $\mathbf{X}_i(\boldsymbol{\theta}_i)$ is an implicit function of the unknown parametric vector $\boldsymbol{\theta}_i$, and $\mathbf{v}_i$ is a measurement noise vector assumed to be a Gaussian white noise vector with zero mean and a covariance matrix $E[\mathbf{v}_i\mathbf{v}_j^T] = \mathbf{R}_i\delta_{ij}$ where δ_{ij} is the Kronecker delta.

The unknown parametric vector $\boldsymbol{\theta}_i$ and the unknown excitation vector $\mathbf{f}_i^*(i = 1, 2, \ldots, k + 1)$ are determined by minimizing an objective function, which is the weighted sum-square error [up to $t = (k + 1)\Delta t$] between the observation data $\mathbf{y}_i$ and the analytical expression $\mathbf{h}$ in Eq. (5), as follows:

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$$J_{k+1} = \sum_{i=1}^{k+1} \mathbf{e}_i^T \mathbf{R}_i^{-1} \mathbf{e}_i; \qquad \mathbf{e}_i = \mathbf{y}_i - \mathbf{h}[\mathbf{X}_i(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i, \mathbf{f}_i^*, \mathbf{f}_i] \tag{6}$$

in which $\mathbf{y}_i$ and $\mathbf{h}[\mathbf{X}_i(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i, \mathbf{f}_i^*, \mathbf{f}_i]$ are given by Eq. (5).

Let $\hat{\theta}_{k+1}$ and $\hat{\mathbf{f}}^*_{k+1|k+1}$ be the estimates of $\boldsymbol{\theta}_{k+1}$ and $\mathbf{f}^*_{k+1}$, respectively, at $t = (k+1)\Delta t$, and $\hat{\mathbf{X}}_{k+1|k}$ be the estimate of $\mathbf{X}_{k+1}$ based on the estimated parametric vector $\hat{\boldsymbol{\theta}}_k$, i.e., $\hat{\mathbf{X}}_{k+1|k} = \mathbf{X}_{k+1}(\hat{\boldsymbol{\theta}}_k)$. Based on the adaptive quadratic sum-square error with unknown inputs (AQSSE-UI) approach described above, the recursive solutions for the estimations $\hat{\boldsymbol{\theta}}_{k+1}$ and $\hat{\mathbf{f}}^*_{k+1|k+1}$ of $\boldsymbol{\theta}_{k+1}$ and $\mathbf{f}^*_{k+1}$, respectively, were derived as follows (Huang et al. 2010):

$$\hat{\mathbf{f}}^*_{k+1|k+1} = \mathbf{S}_{k+1}\mathbf{D}^T_{k+1}\mathbf{R}^{-1}_{k+1}(\mathbf{I} - \mathbf{H}_{k+1}\mathbf{K}_{\theta,k+1}) \times (\mathbf{y}_{k+1} - \hat{\mathbf{h}}_{k+1} + \mathbf{D}_{k+1}\hat{\mathbf{f}}^*_{k|k}) \tag{7}$$

$$\hat{\boldsymbol{\theta}}_{k+1} = \hat{\boldsymbol{\theta}}_k + \mathbf{K}_{\theta,k+1}[\mathbf{y}_{k+1} - \hat{\mathbf{h}}_{k+1} - \mathbf{D}_{k+1}(\hat{\mathbf{f}}^*_{k+1|k+1} - \hat{\mathbf{f}}^*_{k|k})] \tag{8}$$

in which $\mathbf{y}_{k+1}$ is the measured response data and $\mathbf{f}_{k+1}$ is the measured excitation data, respectively, at $t = (k+1)\Delta t$, and

$$\hat{\mathbf{h}}_{k+1} = \mathbf{h}(\hat{\mathbf{X}}_{k+1|k}, \hat{\boldsymbol{\theta}}_k, \hat{\mathbf{f}}^*_{k|k}, \mathbf{f}_{k+1}) \tag{9}$$

$$\mathbf{K}_{\boldsymbol{\theta},k+1} = \mathbf{P}_{\boldsymbol{\theta},k}\mathbf{H}^T_{k+1}(\mathbf{R}_{k+1} + \mathbf{H}_{k+1}\mathbf{P}_{\boldsymbol{\theta},k}\mathbf{H}^T_{k+1})^{-1} \tag{10}$$

$$\mathbf{S}_{k+1} = [\mathbf{D}^T_{k+1}\mathbf{R}^{-1}_{k+1}(\mathbf{I} - \mathbf{H}_{k+1}\mathbf{K}_{\boldsymbol{\theta},k+1})\mathbf{D}_{k+1}]^{-1} \tag{11}$$

$$\mathbf{P}_{\boldsymbol{\theta},k} = (\mathbf{I} + \mathbf{K}_{\boldsymbol{\theta},k}\mathbf{D}_k\mathbf{S}_k\mathbf{D}^T_k\mathbf{R}^{-1}_k\mathbf{H}_k)(\mathbf{I} - \mathbf{K}_{\boldsymbol{\theta},k}\mathbf{H}_k)\mathbf{P}_{\boldsymbol{\theta},k-1} \tag{12}$$

In these equations, $\mathbf{K}_{\boldsymbol{\theta},k+1}$ is the $(n \times l)$ gain matrix for $\hat{\boldsymbol{\theta}}_{k+1}$, $\mathbf{R}_{k+1}$ is the variance matrix of the measurement noises at $t = (k+1)\Delta t$, and $\mathbf{K}_{\boldsymbol{\theta},k}$ and $\mathbf{S}_k$ are obtained, respectively, from Eqs. (10) and (11) by replacing $k+1$ by k. Further, $\mathbf{H}_{k+1}$ and $\mathbf{D}_{k+1}$ in Eqs. (7) and (8) are $(l \times n)$ and $(l \times r)$ matrices given by

$$\mathbf{H}_{k+1} = [\partial \mathbf{h}_{k+1}/\partial \boldsymbol{\theta}_{k+1}]_{\boldsymbol{\theta}_{k+1}=\hat{\boldsymbol{\theta}}_k, \mathbf{f}^*_{k+1}=\hat{\mathbf{f}}^*_{k|k}} \tag{13}$$

$$\mathbf{D}_{k+1} = [\partial \mathbf{h}_{k+1}/\partial \mathbf{f}^*_{k+1}]_{\boldsymbol{\theta}_{k+1}=\hat{\boldsymbol{\theta}}_k, \mathbf{f}^*_{k+1}=\hat{\mathbf{f}}^*_{k|k}} \tag{14}$$

For the numerical computation of the recursive solution above, $\hat{\mathbf{h}}_{k+1} = \mathbf{h}(\hat{\mathbf{X}}_{k+1|k}, \hat{\boldsymbol{\theta}}_k, \hat{\mathbf{f}}^*_{k|k}, \mathbf{f}_{k+1})$ is computed, in which $\hat{\mathbf{X}}_{k+1|k}$ is the estimate of the state vector $\mathbf{X}(t)$ at $t = (k+1)\Delta t$ based on $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}_k$ and $\mathbf{f}^* = \hat{\mathbf{f}}^*_{k|k}$, i.e.,

$$\hat{\mathbf{X}}_{k+1|k} = \hat{\mathbf{X}}_{k|k} + \int_{k\Delta t}^{(k+1)\Delta t} \mathbf{g}(\hat{\mathbf{X}}_{t|k}, \hat{\boldsymbol{\theta}}_k, \hat{\mathbf{f}}^*_{k|k}, \mathbf{f})dt \tag{15}$$

In the original papers for AQSSE and AQSSE-UI (Yang et al. 2009; Huang et al. 2010), the state vector $\hat{\mathbf{X}}_{k+1|k}$ is further up-dated using a Kalman filter. Based on extensive numerical evaluations, this procedure is usually not necessary. Likewise, the computations of $\mathbf{H}_{k+1}$ and $\mathbf{D}_{k+1}$ in Eqs. (13) and (14) can be accomplished using numerical differentiations. It should be mentioned that the recursive solutions in Eqs. (7)–(14) only guarantee the local minimum for the sum-square error, rather than the global minimum. This situation is true for all other damage detection methods based on the recursive solution, such as the extended Kalman filter. Consequently, the estimated parameter values are slightly different if different set of measured data is used.

Experimental Setup for Demonstration of Damage Identification

Recently, experimental tests for a 1/3-scaled six-story steel frame on the shake table have been conducted to demonstrate the capability of a structural damage identification method. These

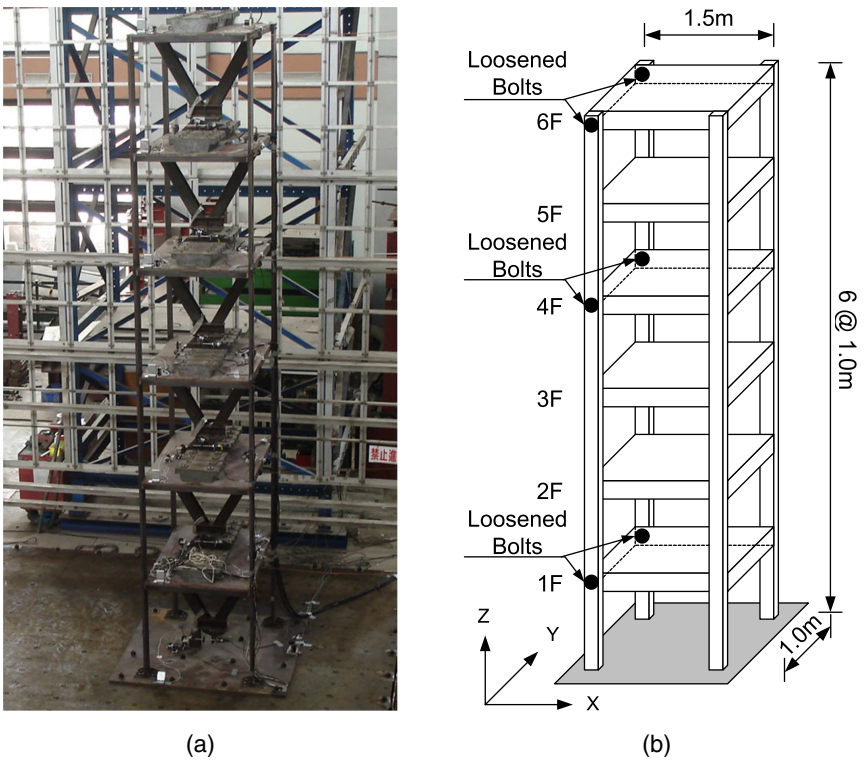


Fig. 1. Six-story steel frame for experimental study: (a) photo of the test structure; (b) sketch showing the dimension of the structure and possible damage locations

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experimental data will be used herein to demonstrate the capability of the proposed damage identification method based on the combination of the reduced-order finite-element model and the AQSSE-UI method.

Experimental Setup of a Six-Story Steel Frame Structure

The test specimen of the six-story steel frame is shown in Figs. 1(a and b), in which the height of each story is 1.0 m and the mass of each floor is 862.85 kg (lumped mass including bracing). The test specimen consists of a single bay with a 1.0-m by 1.5-m floor area. The dimensions are as follows: (1) column: 150×25 mm (rectangular section); (2) beam: $50 \times 50 \times 5$ mm (L-section); (3) floor slab: $1{,}000 \times 1{,}500 \times 20$ mm. The constructions of connections are as follows: (1) the beam-floor is a welded connection, and (2) the column-beam, bracing-floor, and structure-shake table are bolted connections.

The structure is bolted on the shaking table (NCREE shaking table with a dimension of 5×5 m). The undamaged structure was tested using the white noise excitation with a peak acceleration of 50 Gal and also the El Centro earthquake with a peak acceleration of 100 Gal. The time domain excitation and acceleration response signals of each floor were measured. For this six-story steel frame, five cases of joint damage scenarios were artificially introduced by loosening the joint's bolts from their tightened position until a 2-mm gap is produced between the bolt head and the connected member. These five damage scenarios are shown in Fig. 1(b) as black dots.

- Damage scenario I: bolts loosened on 1st floor;
- Damage scenario II: bolts loosened on 1st and 4th floors;
- Damage scenario III: bolts loosened on 4th floor;
- Damage scenario IV: bolts loosened on 4th and 6th floors; and
- Damage scenario V: bolts loosened on 6th floor.

In the damage scenarios above, bolts loosened on a particular floor refers to the situation in which only two joints on that floor are loosened as shown in Fig. 2. This type of damage is a minor damage not critical to the safety of the structure.

Measurements: For all the five cases of damage simulations (loosened bolts), the El Centro earthquake, normalized to a PGA of 100 Gal, was used as the input excitation (along x-direction). In the experimental tests, accelerometers were installed on each floor and on the shake table to measure the acceleration responses of all floors, $\ddot{x}_i$ $(i = 1, 2, \ldots, 6)$, and that of the shake table, $\ddot{x}_0$. The sampling frequency of the measurements is 200 Hz.

For the prediction of the joint damages based on the quadratic sum-square error with unknown inputs (AQSSE-UI), it is not necessary to use all the measurements, i.e., $\ddot{x}_i$ $(i = 1, 2, \ldots, 6)$ and $\ddot{x}_0$. Hence, the damage identification will be carried out using fewer measurements as shown in Table 1. In Table 1, the number of measurements used for the damage identification is designated by the column under *Case index*. Consequently, we shall denote the damage identification cases using both the roman letter (for damage scenarios) and the numeral letter (for the number of measurements used in Table 1). For instance, Case I-7 indicates the damage prediction for the damage scenario I (loosened bolts on the first floor) using seven measurements, i.e., $\ddot{x}_i$ $(i = 1, 2, \ldots, 6)$ and the input excitation $\ddot{x}_0$ (see Table 1). Likewise, Case II-5UI indicates the damage prediction for the damage scenario II (loosened bolts on the first and fourth floors) using five measurements, i.e., $\ddot{x}_1$, $\ddot{x}_2$, $\ddot{x}_4$, $\ddot{x}_5$ and $\ddot{x}_6$, without the input excitation $\ddot{x}_0$ (see Table 1).

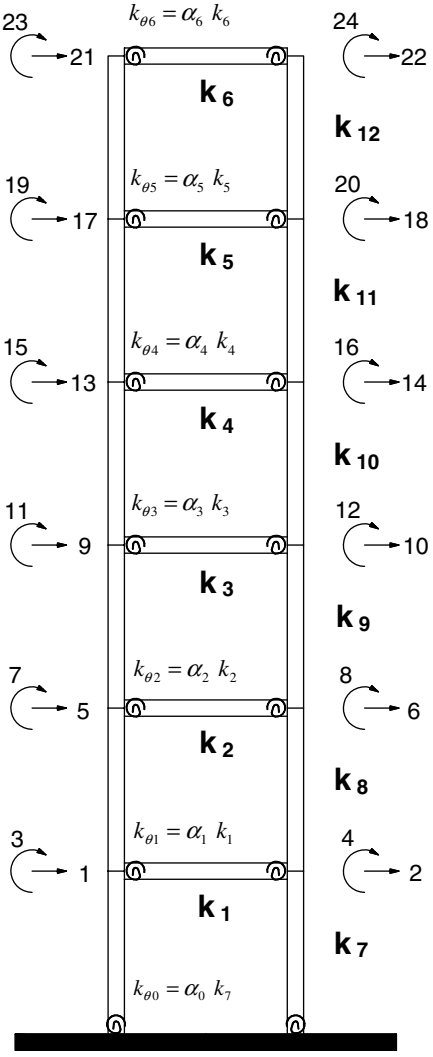


Fig. 2. Finite-element model of the six-story steel frame

Structural Modeling with Semirigid Connection

The original finite element model of the six-story steel frame consists of 18 elements (six beams and 12 columns) with 12 free nodes and seven rotational springs (one pair of symmetric springs on each floor and another pair on the bottom brace connection) as shown in Fig. 2. Since the elements of the frame are steel strips, the axial deformations of the beams and columns are negligible in comparison with the shear and flexural deformations. Hence, each node has two DOF (horizontal and rotational), resulting in a 24-DOF system.

Let E_i, I_i, and L_i be the Young's modulus, moment of inertia and length of the ith element of the structure, respectively, and $k_i = E_i I_i / L_i$ be the equivalent stiffness parameter of the ith member. Further, the joints of the frame are considered as semirigid connections, which are modeled as zero-length rotational springs at both ends of a beam. The rotational stiffness of a beam-to-column

Table 1. Acceleration Measurements Used for Damage Identification

Measurements used	Case index	Measurements used	Case index
$[\ddot{x}_0, \ddot{x}_1, \ddot{x}_2, \ddot{x}_3, \ddot{x}_4, \ddot{x}_5, \ddot{x}_6]$	7	$[\ddot{x}_0, \ddot{x}_2, \ddot{x}_3, \ddot{x}_5]$	4[a]
$[\ddot{x}_0, \ddot{x}_1, \ddot{x}_2, \ddot{x}_4, \ddot{x}_5, \ddot{x}_6]$	6	$[\ddot{x}_1, \ddot{x}_2, \ddot{x}_3, \ddot{x}_4, \ddot{x}_5, \ddot{x}_6]$	6UI
$[\ddot{x}_0, \ddot{x}_1, \ddot{x}_2, \ddot{x}_3, \ddot{x}_5, \ddot{x}_6]$	6[a]	$[\ddot{x}_1, \ddot{x}_2, \ddot{x}_4, \ddot{x}_5, \ddot{x}_6]$	5UI
$[\ddot{x}_0, \ddot{x}_1, \ddot{x}_2, \ddot{x}_4, \ddot{x}_6]$	5	$[\ddot{x}_1, \ddot{x}_2, \ddot{x}_3, \ddot{x}_5, \ddot{x}_6]$	5UI[a]
$[\ddot{x}_0, \ddot{x}_2, \ddot{x}_3, \ddot{x}_5, \ddot{x}_6]$	5[a]	$[\ddot{x}_1, \ddot{x}_2, \ddot{x}_4, \ddot{x}_6]$	4UI
$[\ddot{x}_0, \ddot{x}_1, \ddot{x}_4, \ddot{x}_6]$	4	$[\ddot{x}_2, \ddot{x}_3, \ddot{x}_5, \ddot{x}_6]$	4UI[a]

[a]Measurement sets with the same number of measurements. For example, case index 6 indicates the measurement set of $[\ddot{x}_0, \ddot{x}_1, \ddot{x}_2, \ddot{x}_4, \ddot{x}_5, \ddot{x}_6]$, but case index 6[a] indicates $[\ddot{x}_0, \ddot{x}_1, \ddot{x}_2, \ddot{x}_3, \ddot{x}_5, \ddot{x}_6]$.

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connection $k_{\theta i}$ is expressed as $k_{\theta i} = \alpha_i EI/L$, where α_i is the rotational stiffness factor and EI/L is the corresponding element stiffness. Hence, the joint damages (or loosened bolts) can be identified through the reduction of the rotational stiffness factor α_i. Further, Rayleigh damping is assumed, where α and β are the mass and stiffness-proportional damping coefficients. Thus, the construction of the stiffness matrix taking into account the rotational joint stiffness for the frame in Fig. 2 can be assembled. Finally, the 24-DOF system was condensed to a six-DOF reduced-order system (see Table 1) for the damage identification purpose using the static condensation method (Chopra 2001).

Damage Identification Based on Experimental Data

As mentioned previously, the undamaged structure was tested using white-noise excitation and also the El Centro earthquake. The unknown parameters of the undamaged structure are almost identical based on the white nose excitation and the El Centro earthquake. This has been expected because the unknown parameters are independent of input excitations. Then, five damage scenarios were artificially introduced by loosening the bolts in joints on different floors. The damaged frame was then subject to 30 s of the El Centro earthquake scaled to a PGA of 100 Gal. Acceleration response quantities of each floor and of the shake table were recorded. Different response quantities will be used for the damage identification (see Table 1) in this section.

Unknown Structural Parameters and Initial Values

The six-story frame is a symmetric structure and a total of 21 unknown structural parameters will be identified. The unknown parametric vector is given by $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_{21}]^T = [k_1, k_2, k_3, k_4, k_5, k_6, k_7, k_8, k_9, k_{10}, k_{11}, k_{12}, \alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha, \beta]^T$, where $\theta_1 \sim \theta_6 = k_1 \sim k_6$ are the stiffness parameters of 6 beam elements, $\theta_7 \sim \theta_{12} = k_7 \sim k_{12}$ are the stiffness parameters of six column elements, $\theta_{13} = \alpha_0$ and $\theta_{14} \sim \theta_{19} = \alpha_1 \sim \alpha_6$ are rotational stiffness parameters of the column-shake table and column-beam connections, and $\theta_{20} = \alpha$ and $\theta_{21} = \beta$ are, respectively, the mass and stiffness-proportional Rayleigh damping coefficients (see Fig. 2).

To start the recursive solution in Eqs. (7) and (8), the initial values should be assigned reasonably, e.g., damping and stiffness should be positive values. The initial values of state variables are zero. The assigned initial values for k_i $(i = 1, 2, \ldots, 12)$, α_i $(i = 0, 1, \ldots, 6)$, α and β are: $k_1 = k_2 = \cdots = k_6 = 6.0 \times 10^5$ N · m, $k_7 = k_8 = \cdots = k_{12} = 6.0 \times 10^4$ N · m, $\alpha_0 = 8$ rad^{-1}, $\alpha_1 = \alpha_2 = \cdots = \alpha_6 = 0.5$ rad^{-1}, $\alpha = 1 \times 10^{-1}$ s^{-1} and $\beta = 1 \times 10^{-4}$ s, respectively. In the recursive solutions, all the parameters have been scaled to the same order of magnitude (in the order of 10) for the convenience of computations. The initial error covariance matrix of the parametric vector is chosen to be $\mathbf{P}_{\boldsymbol{\theta},0} = 10\mathbf{I}_{21}$ and the initial sensitivity matrix is zero, i.e., $\mathbf{X}_{\boldsymbol{\theta}} = \mathbf{0}_{12\times 21}$. The covariance matrix of the measurement noise vector $\mathbf{v}(t)$ is chosen to be $\mathbf{R} = \mathbf{I}_6$, where $\mathbf{I}_j$ is a $(j \times j)$ identity matrix.

All the system identification methods involving the recursive solutions, such as the extended Kalman filter and AQSSE-UI, require the initial guesses to start with. The convergence to the final solutions depends on how far the initial guesses are. For the AQSSE-UI, the convergence is very fast if the initial guesses are of the same order of magnitude of the converged values. The convergence process is slower if the initial guesses are further apart from the converged values. Based on our numerical results, given the same initial guesses, the AQSSE-UI technique converges much faster than any other techniques (Huang et al. 2010).

Identification of Undamaged Frame

Based on the reduced-order finite-element model (six DOF) and the AQSSE-UI technique, the identified unknown structural parameters

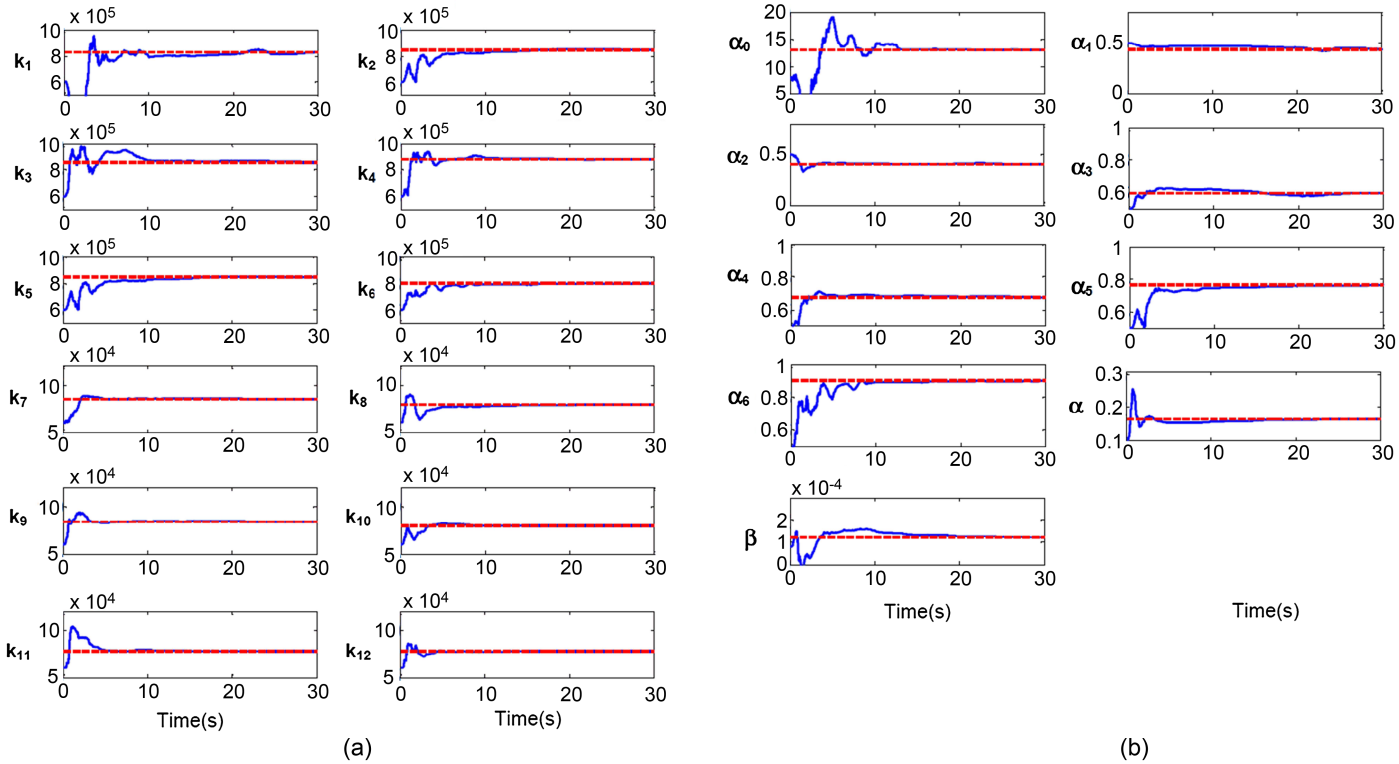


Fig. 3. (a) Identified parameters $k_1 \sim k_{12}$ (in N · m); (b) identified parameters $\alpha_0 \sim \alpha_6$ (in rad^{-1}), α (s^{-1}), and β (s)

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Table 2. Reference Values Used for Damage Identification

Structural parameter	Reference value	Structural parameter	Reference value	Structural parameter	Reference value	Structural parameter	Reference value
k_1	8.48×10^5	k_7	8.81×10^4	α_1	0.42	α_0	12.6
k_2	7.68×10^5	k_8	7.76×10^4	α_2	0.41	α	1.66
k_3	8.70×10^5	k_9	8.62×10^4	α_3	0.58	β	1.12
k_4	8.54×10^5	k_{10}	7.83×10^4	α_4	0.69		
k_5	8.67×10^5	k_{11}	7.89×10^4	α_5	0.75		
k_6	7.92×10^5	k_{12}	7.70×10^4	α_6	0.89		

Note: Reference value (k_i in N · m, α_i in rad^{-1}, α in 10^{-1} s^{-1}, and β in 10^{-4} s).

for the undamaged case, using all the measured records, i.e., six floor acceleration responses and the acceleration of the shake table, are presented in Fig. 3 as solid curves. Also shown as dashed lines in Fig. 3 are the finally converged values, referred to as the *reference values*. These reference values are shown in Table 2. It is observed from Fig. 3 that the unknown structural parameters converge nicely.

Further, the identified acceleration responses of all floors are shown in Fig. 4 as solid curves, whereas the corresponding experimental accelerations are shown in the same figure as dashed curves for comparison. In order to magnify the comparison, only a duration of 20 s is plotted in Fig. 4. As observed from Fig. 4, the identified acceleration responses correlated well with that of the recorded responses.

Identification of Damaged Frame

In a similar manner, the identification for unknown structural parameters θ_i ($i = 1, 2, \ldots, 21$) can be made for different damage scenarios using different numbers of measurements (Table 1). Fig. 5 shows the identified results θ_i for five damage scenarios (I-V) using seven records, including acceleration responses of six floors and the acceleration of the shake table. These results are denoted by Cases I-7 to V-7. The bar chart shown in Fig. 5 displays the identified results normalized by the reference values presented in Table 2 for the undamaged case. Note that α_1, α_4, and α_6 are the rotational stiffness parameters of the joints on the first, fourth and sixth floors, respectively (see Fig. 2). Fig. 5 clearly demonstrates that the proposed damage identification method is capable of identifying the damage locations of bolted joints.

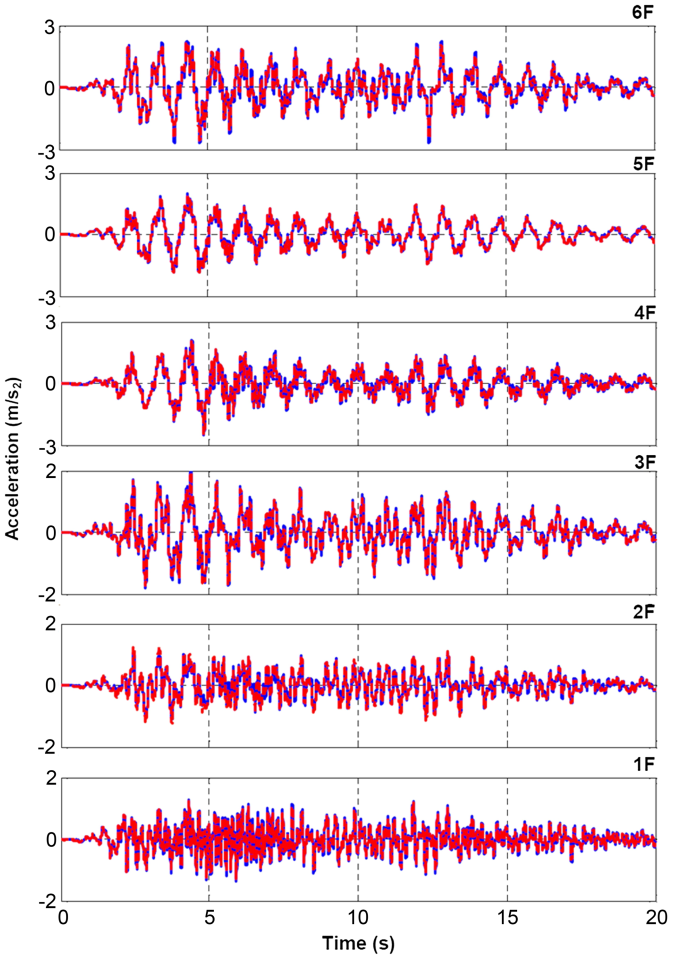


Fig. 4. Comparison between the identified acceleration responses and the measured responses

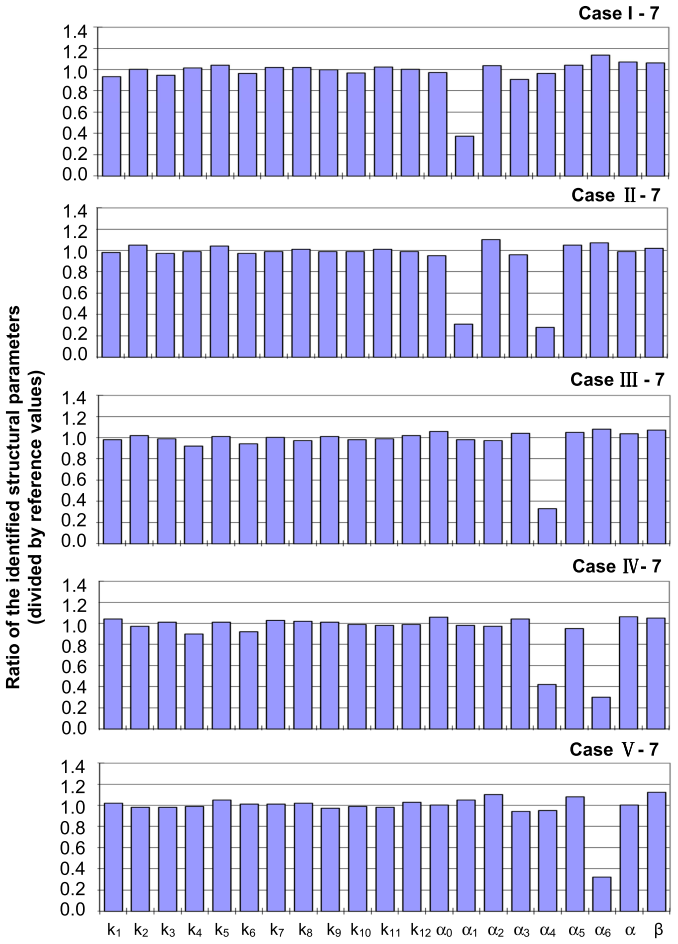


Fig. 5. Ratios of identified structural parameters for five damage scenarios (Cases I-7 to V-7)

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Fig. 6 shows the identified results for the unknown structural parameters θ_i $(i = 1, 2, \ldots, 21)$ for five damage scenarios using only acceleration measurements of five floors, i.e., $\ddot{x}_1$, $\ddot{x}_2$, $\ddot{x}_4$, $\ddot{x}_5$ and $\ddot{x}_6$, without the shake table excitation $\ddot{x}_0$. The results are normalized using undamaged system generated under the same instrument array, i.e., five acceleration measurements. These results are denoted by Cases I-5UI to V-5UI. As observed from Fig. 6, the proposed damage identification method is capable of detecting the location of damaged joints in the frame. The identified unknown shake table excitation $\ddot{x}_0$ for the damage scenario II (Case II-5UI) is shown as solid curves in Fig. 7(a) from 3 to 6 s and in Fig. 7(b) from 10 to 15 s. For comparison, also shown in these figures, as dashed curves, are the measured shake table excitations. Again, different time scales are presented in Fig. 7 in order to magnify the comparison. Fig. 7 clearly demonstrates that the identified unknown shake table excitation correlates well with the measured records.

Further, the identified results for the unknown structural parameters θ_i $(i = 1, 2, \ldots, 21)$ for five damage scenarios using only acceleration measurements of four floors, i.e., $\ddot{x}_1$, $\ddot{x}_2$, $\ddot{x}_4$, and $\ddot{x}_6$, without the shake table excitation $\ddot{x}_0$ are presented in Fig. 8, denoted by Cases I-4UI to V-4UI. The results are normalized using undamaged system generated under the same instrument array, i.e., four acceleration measurements. Again, Fig. 8 demonstrates that the proposed damage identification method is capable of detecting the location of damaged bolt connections. Numerical results for different identification cases shown in Table 1 have been obtained; however, these results are not presented, since they are similar to Figs. 6–8. Furthermore, a different damage scenario corresponds to a different global minimum J_{k+1} in Eq. (6). However, the recursive solutions obtained in Eqs. (7) and (8) usually correspond to a local minimum. As a result, ratios of solutions presented in Figs. 5, 6, and 8 are either above or below 1.0. The problem of searching for the global minimum solution is a subject of future research.

In this paper, the ratio of the stiffness in the undamaged system to that in the damaged system, referred to as the damage sensitivity feature (DSF), is used to indicate the damage. It is important to set a threshold value below which the damage is signified. However, such a threshold value depends on the severity of the damage to be detected. For the detection of minor damages in the joint connection considered in this paper, a threshold value of DSF equal to 0.5 will be sufficient (see Figs. 5, 6, and 8).

The AQSSE-UI damage identification method used herein is capable of identifying structural damages, including the locations of the joint damages in the frame, using only a limited number of sensor measurements. However, as the number of response measurements reduces, the accuracy of the damage identification degrades. The question of the minimum number of measurements required to achieve a certain level of accuracy for the damage detection remains to be studied, which is a very challenging problem. Nevertheless, the proposed method, consisting of the reduced-order

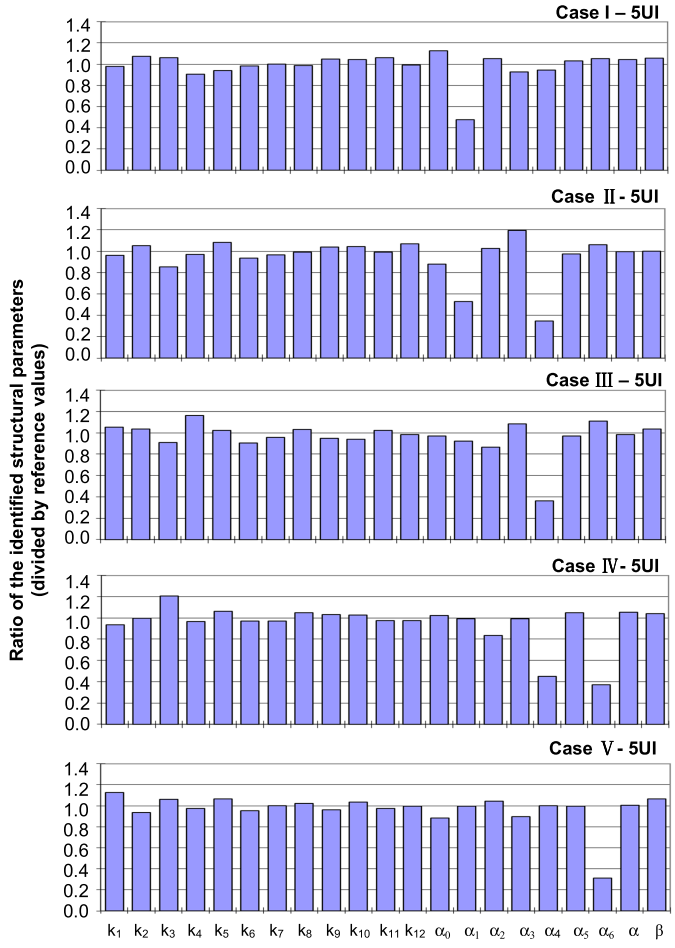


Fig. 6. Ratio of identified structural parameters for five damage scenarios (Cases I-5UI to V-5UI)

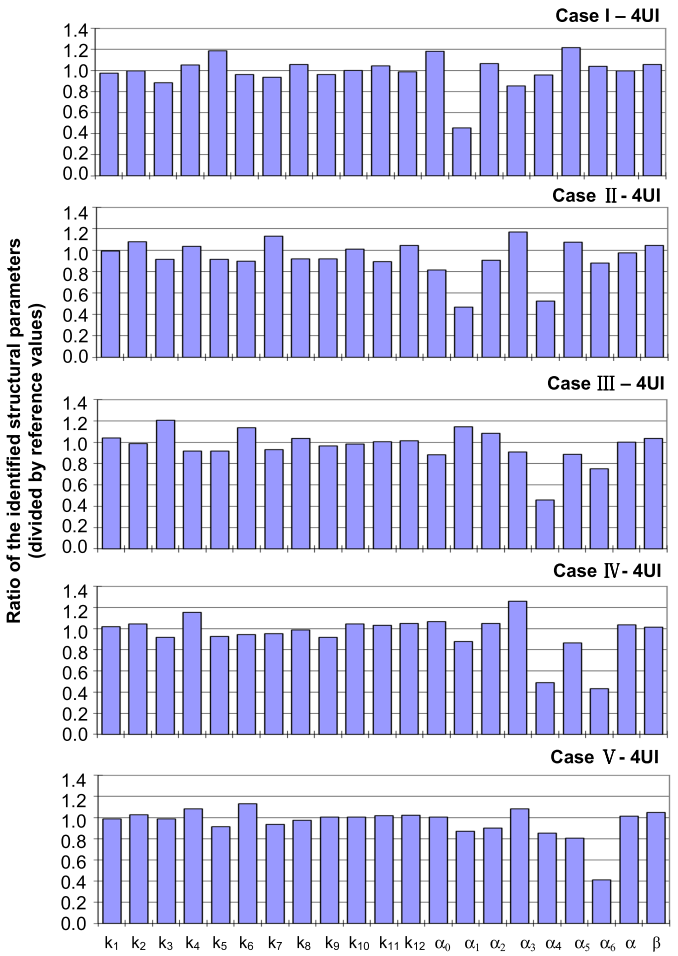


Fig. 7. Identified unknown excitation (Case II-5UI)

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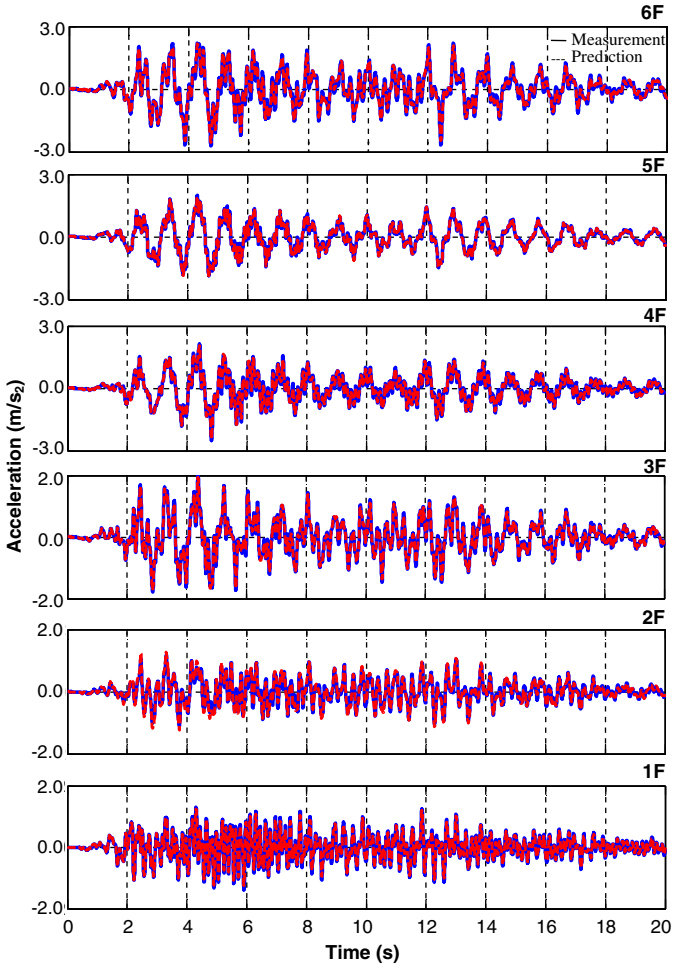


Fig. 8. Ratio of identified structural parameters for five damage scenarios (Cases I-4UI to V-4UI)

finite-element formulation and the AQSSE-UI technique, reduces a significant number of sensor measurements as demonstrated by the experimental results.

Conclusion

Based on a combination of the reduced-order finite-element model and the adaptive quadratic sum-square error with unknown inputs, a method has been presented to identify system parameters and unknown excitations for complex structures. The method of static condensation has been used to reduce the number of degrees of freedom of the finite-element model, whereas the same number of unknown structural parameters in the original complex structure has been retained in the reduced-order system. This approach not only reduces the number of required sensor measurements but also removes response quantities which are difficult to measure, such as the rotational acceleration of a nodal point. Likewise, the AQSSE-UI technique is capable of identifying unknown structural parameters using only a limited number of measurements. Shake table experimental tests using a six-story steel frame have been conducted at NCREE, in which various damage scenarios in terms of loosened bolt joints have been introduced. Experimental data demonstrate that the proposed damage identification method is capable of identifying loosened bolts in the joints of the steel frame.

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